Comparison for W3 and 13

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```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(purrr)
library(stringr)
library(ggplot2)
library(clue)
```

```r
m_w3_nocov  <- readRDS("lca_w3_k4_nocov.rds")
m_w13_nocov <- readRDS("lca_w13_k4_nocov_update.rds")
m_w3_cov    <- readRDS("lca_w3_k4_cov.rds")
m_w13_cov   <- readRDS("lca_w13_k4_cov_update.rds")
```

```r
# Fit comparison (AIC/BIC/G^2, N, K)
fit_row <- function(m, wave, cov){
  tibble(
    wave     = wave,
    covars   = cov,
    k        = ncol(m$P),
    N        = m$N,
    AIC      = m$aic,
    BIC      = m$bic,
    G2       = m$Gsq
  )
}
fit_cmp <- bind_rows(
  fit_row(m_w3_nocov,  "W3",  "no"),
  fit_row(m_w13_nocov, "W13", "no"),
  fit_row(m_w3_cov,    "W3",  "yes"),
  fit_row(m_w13_cov,   "W13", "yes")
)
fit_cmp
```

```
## # A tibble: 4 x 6
```

```
##   wave  covars     N    AIC    BIC    G2
##   <chr> <chr>  <int>  <dbl>  <dbl> <dbl>
## 1 W3    no     16351 61044. 61314.  289.
## 2 W13   no     13700 53943. 54206.  313.
## 3 W3    yes    16351 58226. 58911.  686.
## 4 W13   yes    13700 51149. 51819.  583.
```

```r
# Class size comparison (estimated mixing proportions)
mix_row <- function(m, wave, cov){
  tibble(wave = wave, covars = cov, class = seq_along(m$P), proportion = as.numeric(m$P))
}
mix_cmp <- bind_rows(
  mix_row(m_w3_nocov,  "W3",  "no"),
  mix_row(m_w13_nocov, "W13", "no"),
  mix_row(m_w3_cov,    "W3",  "yes"),
  mix_row(m_w13_cov,   "W13", "yes")
)
mix_cmp
```

```
## # A tibble: 16 x 4
##    wave  covars class proportion
##    <chr> <chr>  <int>      <dbl>
##  1 W3    no         1     0.0147
##  2 W3    no         2     0.797
##  3 W3    no         3     0.149
##  4 W3    no         4     0.0385
##  5 W13   no         1     0.166
##  6 W13   no         2     0.0183
##  7 W13   no         3     0.777
##  8 W13   no         4     0.0379
##  9 W3    yes        1     0.0239
## 10 W3    yes        2     0.757
## 11 W3    yes        3     0.0275
## 12 W3    yes        4     0.192
## 13 W13   yes        1     0.270
## 14 W13   yes        2     0.0481
## 15 W13   yes        3     0.661
## 16 W13   yes        4     0.0208
```

```r
# Quick entropy / classification sharpness
entropy <- function(p) -sum(p * log(p + 1e-12))
norm_entropy <- function(post) {
  # average normalized entropy across cases (0 = sharp, 1 = diffuse)
  K <- ncol(post); Hmax <- log(K)
  mean(apply(post, 1, entropy) / Hmax)
}
sharpness <- function(post) mean(apply(post, 1, max))  # closer to 1 is better

qual_cmp <- tibble(
  wave   = c("W3","W13","W3","W13"),
  covars = c("no","no","yes","yes"),
  norm_entropy = c(norm_entropy(m_w3_nocov$posterior),
                   norm_entropy(m_w13_nocov$posterior),
                   norm_entropy(m_w3_cov$posterior),
                   norm_entropy(m_w13_cov$posterior)),
```

```r
  max_p_mean   = c(sharpness(m_w3_nocov$posterior),
                   sharpness(m_w13_nocov$posterior),
                   sharpness(m_w3_cov$posterior),
                   sharpness(m_w13_cov$posterior))
)
qual_cmp
```

```
## # A tibble: 4 x 4
##   wave  covars norm_entropy max_p_mean
##   <chr> <chr>         <dbl>      <dbl>
## 1 W3    no            0.215      0.891
## 2 W13   no            0.245      0.867
## 3 W3    yes           0.196      0.896
## 4 W13   yes           0.212      0.881
```

We evaluated classification quality using normalized entropy (lower = better) and the mean maximum posterior probability (higher = better).

For Wave 3, the no-covariate model showed clearer class separation than the covariate model (entropy: 0.215 vs 0.308; max posterior: 0.891 vs 0.851). This suggests that covariates introduced additional variability and slightly diluted classification certainty. In contrast, for Wave 13, adding covariates improved classification quality (entropy: 0.245 to 0.207; max posterior: 0.867 to 0.886), indicating that class membership in Wave 13 is more strongly structured by the covariates.

Do the proportions of respondents in each class differ between 1996 and 2016?

Without covariances

```r
match_classes <- function(m_w3, m_w13){

  flatten_probs <- function(m){
    mats <- lapply(m$probs, function(x) as.matrix(x))
    P <- do.call(cbind, mats)
    rownames(P) <- paste0("c", seq_len(nrow(P)))
    P
  }

  # get flattened matrices
  P_w3  <- flatten_probs(m_w3)
  P_w13 <- flatten_probs(m_w13)

  # cosine similarity
  cos_sim <- function(A, B){
    A <- as.matrix(A); B <- as.matrix(B)
    num <- A %*% t(B)
    A_n <- sqrt(rowSums(A^2)); B_n <- sqrt(rowSums(B^2))
    sweep(sweep(num, 1, A_n, "/"), 2, B_n, "/")
  }

  S <- cos_sim(P_w3, P_w13)
  S[is.na(S)] <- 0

  # Hungarian: convert similarity to cost
  cost <- max(S) - S
  assign <- clue::solve_LSAP(cost)
```

```r
  matched_pairs <- tibble(
    w3_class   = seq_len(nrow(S)),
    w13_class  = as.integer(assign),
    similarity = S[cbind(seq_len(nrow(S)), as.integer(assign))]
  )

  return(matched_pairs)
}
```

```r
compare_props <- function(matched_pairs, m_w3, m_w13){
  p_w3 <- tibble(
    w3_class = seq_along(m_w3$P),
    prop_w3  = as.numeric(m_w3$P)
  )

  p_w13 <- tibble(
    w13_class = seq_along(m_w13$P),
    prop_w13  = as.numeric(m_w13$P)
  )

  matched_pairs %>%
    left_join(p_w3,  by = "w3_class") %>%
    left_join(p_w13, by = "w13_class")
}
```

```r
# no covariates
matched_nocov <- match_classes(m_w3_nocov, m_w13_nocov)
prop_nocov    <- compare_props(matched_nocov, m_w3_nocov, m_w13_nocov)

prop_nocov
```

```
## # A tibble: 4 x 5
##   w3_class w13_class similarity prop_w3 prop_w13
##      <int>     <int>      <dbl>   <dbl>    <dbl>
## 1        1         2      0.986  0.0147   0.0183
## 2        2         3      0.999  0.797    0.777
## 3        3         1      0.993  0.149    0.166
## 4        4         4      0.971  0.0385   0.0379
```

To compare latent classes across waves, we aligned Wave 3 and Wave 13 classes based on cosine similarity of their item-response probability profiles.

High similarity indicates that the classes represent very similar underlying subpopulations.

With covariance

```r
# with covariates
matched_cov <- match_classes(m_w3_cov, m_w13_cov)
prop_cov    <- compare_props(matched_cov, m_w3_cov, m_w13_cov)

prop_cov
```

```
## # A tibble: 4 x 5
##   w3_class w13_class similarity prop_w3 prop_w13
##      <int>     <int>      <dbl>   <dbl>    <dbl>
## 1        1         1      0.991  0.0239   0.270
```

```
## 2        2         3      1.00   0.757    0.661
## 3        3         2      0.982  0.0275   0.0481
## 4        4         4      0.952  0.192    0.0208
```

Which demographic covariates predict class membership, and do these associations differ across years?

```
m_w3_cov$coeff
```

```
##                            [,1]          [,2]          [,3]
## (Intercept)        0.9815236110 -0.7236228364  1.1985445824
## age               -0.0014315012  0.0005313769 -0.0007589214
## factor(gender)2    0.8077256954  0.7970673262  1.2167073328
## factor(degree)1    0.9787203159 -0.4441534759 -0.5331295437
## factor(degree)2    2.9173487920  0.9220569097  1.0985276858
## factor(degree)3    2.8629663144 -2.5271436058 -0.3530719937
## factor(degree)4    1.3584654802 -0.3498166864 -2.3267509033
## factor(degree)5    2.6332695825 -2.4420742573 -0.3131063297
## factor(degree)6    2.2838056937 -2.2507122354  0.0268857669
## factor(degree)9    0.0057578004 -0.0108508164  0.0106959353
## factor(race)1      0.9959389532 -0.5812313459 -0.1509554774
## factor(race)2      0.0712908861  0.6904060912  1.3638514984
## factor(race)7     -0.0933196248 -0.8059918239 -0.0163351193
## factor(usborn)5   -1.9938112084 -0.9499049047 -0.9311669447
## factor(usborn)9   -0.0003085191 -0.0463364801  0.0343398354
## factor(hispanic)1 -1.7434272576 -0.8366692916 -1.4098535657
## factor(hispanic)2  0.3026604724 -1.4496418050  0.8686982135
## factor(hispanic)3  0.1735255648 -0.1178445205  0.0132015477
## factor(hispanic)5  2.2435780898  1.7201285090  1.7184404053
```

```
m_w13_cov$coeff
```

```
##                          [,1]        [,2]         [,3]
## (Intercept)       -10.64105331 -6.75778263   3.96828014
## age                 0.12845490  0.01317688  -0.18335298
## factor(gender)2    -0.43957080 -0.32189245   3.37310491
## factor(degree)1     0.07002859  1.12917089  -0.03915765
## factor(degree)2     0.15999215  1.85934504  -3.84353709
## factor(degree)3     0.88914555  2.70580882  12.18694832
## factor(degree)4     1.40872395  3.71276216   2.91335281
## factor(degree)5     1.84696161  4.12755432   3.60086156
## factor(degree)6     1.39433263  3.91239227   5.52621947
## factor(degree)9     1.04378630  2.80263171  -3.00038410
## factor(race)1       1.70612364  1.98580092  -3.59062659
## factor(race)2       1.20649503  0.55067870  -5.65048559
## factor(race)7       1.54693823  0.93100277 -13.54308284
## factor(usborn)5    -0.95264829 -1.68815023  -9.90995576
## factor(usborn)9    -3.73785896 -2.82841007   1.73911277
## factor(hispanic)1  -1.78348242  2.95285075  -8.13449262
## factor(hispanic)2  -2.12236794  3.04284335   1.72416285
## factor(hispanic)3  -9.50667168  4.33321691   4.50546503
## factor(hispanic)5  -1.90136186  4.06512414  -0.67129086
```

```
m_w3_cov$probs
```

```
## $Month
##                  Pr(1)     Pr(2)
```

```
## class 1:  0.040284838 0.9597152
## class 2:  0.009834396 0.9901656
## class 3:  0.459750026 0.5402500
## class 4:  0.023788171 0.9762118
##
## $Day
##               Pr(1)     Pr(2)
## class 1:  0.2076370 0.7923630
## class 2:  0.1215710 0.8784290
## class 3:  0.8266226 0.1733774
## class 4:  0.2365176 0.7634824
##
## $Year
##                 Pr(1)     Pr(2)
## class 1:  0.034695296 0.9653047
## class 2:  0.003684503 0.9963155
## class 3:  0.623008825 0.3769912
## class 4:  0.020527430 0.9794726
##
## $DOW
##                 Pr(1)     Pr(2)
## class 1:  0.003814347 0.9961857
## class 2:  0.010267782 0.9897322
## class 3:  0.403048288 0.5969517
## class 4:  0.024872228 0.9751278
##
## $Scissors
##                 Pr(1)     Pr(2)
## class 1:  0.011061903 0.9889381
## class 2:  0.005955512 0.9940445
## class 3:  0.083284318 0.9167157
## class 4:  0.024406736 0.9755933
##
## $Cactus
##                Pr(1)     Pr(2)
## class 1:  0.03678652 0.9632135
## class 2:  0.01686188 0.9831381
## class 3:  0.50762608 0.4923739
## class 4:  0.40962553 0.5903745
##
## $President
##               Pr(1)     Pr(2)
## class 1:  0.1688190 0.8311810
## class 2:  0.0138676 0.9861324
## class 3:  0.5382075 0.4617925
## class 4:  0.1720836 0.8279164
##
## $VP
##               Pr(1)     Pr(2)
## class 1:  0.8979558 0.1020442
## class 2:  0.1527047 0.8472953
## class 3:  0.8506572 0.1493428
## class 4:  0.6701370 0.3298630
```

```r
m_w13_cov$probs
```

```
## $Month
##                Pr(1)     Pr(2)
## class 1:  0.02805527 0.9719447
## class 2:  0.49464279 0.5053572
## class 3:  0.01456642 0.9854336
## class 4:  0.01852626 0.9814737
##
## $Day
##               Pr(1)     Pr(2)
## class 1:  0.2002143 0.7997857
## class 2:  0.8576229 0.1423771
## class 3:  0.1446663 0.8553337
## class 4:  0.0485432 0.9514568
##
## $Year
##                 Pr(1)     Pr(2)
## class 1:  0.029873870 0.9701261
## class 2:  0.584536638 0.4154634
## class 3:  0.008565042 0.9914350
## class 4:  0.009279287 0.9907207
##
## $DOW
##                Pr(1)     Pr(2)
## class 1:  0.04152165 0.9584783
## class 2:  0.44409802 0.5559020
## class 3:  0.01449243 0.9855076
## class 4:  0.01197027 0.9880297
##
## $Scissors
##                  Pr(1)     Pr(2)
## class 1:  3.246430e-02 0.9675357
## class 2:  6.582248e-02 0.9341775
## class 3:  4.577773e-03 0.9954222
## class 4:  8.645077e-17 1.0000000
##
## $Cactus
##                  Pr(1)     Pr(2)
## class 1:  2.197480e-01 0.7802520
## class 2:  2.890317e-01 0.7109683
## class 3:  9.838201e-03 0.9901618
## class 4:  5.817547e-49 1.0000000
##
## $President
##                    Pr(1)     Pr(2)
## class 1:   3.772806e-02 0.9622719
## class 2:   3.503764e-01 0.6496236
## class 3:   1.704920e-03 0.9982951
## class 4:  1.314356e-116 1.0000000
##
## $VP
##               Pr(1)     Pr(2)
## class 1:  0.7840031 0.2159969
```

```
## class 2:  0.8483892 0.1516108
## class 3:  0.1671725 0.8328275
## class 4:  0.3266551 0.6733449
```

```
m_w3_nocov$probs
```

```
## $Month
##                   Pr(1)     Pr(2)
## class 1:  0.633396914 0.3666031
## class 2:  0.005508788 0.9944912
## class 3:  0.029248199 0.9707518
## class 4:  0.195826123 0.8041739
##
## $Day
##                 Pr(1)      Pr(2)
## class 1:  0.9003198 0.09968016
## class 2:  0.1057820 0.89421798
## class 3:  0.2799834 0.72001660
## class 4:  0.6654896 0.33451038
##
## $Year
##                   Pr(1)     Pr(2)
## class 1:  0.828735546 0.1712645
## class 2:  0.001891506 0.9981085
## class 3:  0.036864284 0.9631357
## class 4:  0.142788567 0.8572114
##
## $DOW
##                   Pr(1)     Pr(2)
## class 1:  0.464695278 0.5353047
## class 2:  0.004776264 0.9952237
## class 3:  0.031208886 0.9687911
## class 4:  0.218567216 0.7814328
##
## $Scissors
##                   Pr(1)     Pr(2)
## class 1:  0.086456894 0.9135431
## class 2:  0.006483089 0.9935169
## class 3:  0.026353363 0.9736466
## class 4:  0.035395135 0.9646049
##
## $Cactus
##                  Pr(1)     Pr(2)
## class 1:  0.62110589 0.3788941
## class 2:  0.04907937 0.9509206
## class 3:  0.33994877 0.6600512
## class 4:  0.18292733 0.8170727
##
## $President
##                    Pr(1)     Pr(2)
## class 1:  7.079249e-01 0.2920751
## class 2:  1.089080e-02 0.9891092
## class 3:  2.893035e-01 0.7106965
## class 4:  3.740145e-12 1.0000000
##
```

```
## $VP
##                Pr(1)      Pr(2)
## class 1:  0.9538861 0.04611392
## class 2:  0.1626035 0.83739647
## class 3:  0.8974073 0.10259266
## class 4:  0.2877396 0.71226043
```

```
m_w13_nocov$probs
```

```
## $Month
##                 Pr(1)     Pr(2)
## class 1:  0.03557391 0.9644261
## class 2:  0.70425005 0.2957500
## class 3:  0.01021639 0.9897836
## class 4:  0.38540887 0.6145911
##
## $Day
##               Pr(1)     Pr(2)
## class 1:  0.2830912 0.7169088
## class 2:  0.9322466 0.0677534
## class 3:  0.1261457 0.8738543
## class 4:  0.7831921 0.2168079
##
## $Year
##                 Pr(1)     Pr(2)
## class 1:  0.04630804 0.9536920
## class 2:  0.86246093 0.1375391
## class 3:  0.00694968 0.9930503
## class 4:  0.34613407 0.6538659
##
## $DOW
##                 Pr(1)     Pr(2)
## class 1:  0.06238320 0.9376168
## class 2:  0.63709138 0.3629086
## class 3:  0.01129816 0.9887018
## class 4:  0.30494992 0.6950501
##
## $Scissors
##                  Pr(1)     Pr(2)
## class 1:  0.047630753 0.9523692
## class 2:  0.111216151 0.8887838
## class 3:  0.005780881 0.9942191
## class 4:  0.013239502 0.9867605
##
## $Cactus
##                 Pr(1)     Pr(2)
## class 1:  0.29691797 0.7030820
## class 2:  0.45806403 0.5419360
## class 3:  0.02525369 0.9747463
## class 4:  0.06125750 0.9387425
##
## $President
##                   Pr(1)     Pr(2)
## class 1:  9.530929e-02 0.9046907
## class 2:  6.064717e-01 0.3935283
```

```
## class 3:  5.076530e-16 1.0000000
## class 4:  3.149930e-02 0.9685007
##
## $VP
##               Pr(1)        Pr(2)
## class 1:  0.8702294 1.297706e-01
## class 2:  1.0000000 5.162249e-31
## class 3:  0.2405931 7.594069e-01
## class 4:  0.5196675 4.803325e-01
```

How can we assess the model fit using normalized entropy and the average maximum posterior probability?

```
entropy_row <- function(p_row){
  -sum(p_row * log(p_row + 1e-12))
}

norm_entropy_model <- function(post_mat){
  K <- ncol(post_mat)
  Hmax <- log(K)
  H_i <- apply(post_mat, 1, entropy_row)
  mean(H_i / Hmax)
}

sharpness_model <- function(post_mat){
  mean(apply(post_mat, 1, max))
}

class_quality <- function(mod, wave, covars){
  post <- mod$posterior
  tibble(
    wave         = wave,
    covars       = covars,   # "yes" / "no"
    n            = nrow(post),
    K            = ncol(post),
    norm_entropy = norm_entropy_model(post),
    max_p_mean   = sharpness_model(post)
  )
}
```

```
qual_cmp <- bind_rows(
  class_quality(m_w3_nocov,  wave = "W3",  covars = "no"),
  class_quality(m_w13_nocov, wave = "W13", covars = "no"),
  class_quality(m_w3_cov,    wave = "W3",  covars = "yes"),
  class_quality(m_w13_cov,   wave = "W13", covars = "yes")
)

qual_cmp
```

```
## # A tibble: 4 x 6
##   wave  covars     n     K norm_entropy max_p_mean
##   <chr> <chr>  <int> <int>        <dbl>      <dbl>
## 1 W3    no     16351     4        0.215      0.891
## 2 W13   no     13700     4        0.245      0.867
## 3 W3    yes    16351     4        0.196      0.896
```

```
## 4 W13    yes     13700      4         0.212       0.881
```

```r
library(ggplot2)

ggplot(qual_cmp, aes(x = wave, y = norm_entropy,
                     group = covars, linetype = covars)) +
  geom_line() +
  geom_point() +
  labs(x = "Wave",
       y = "Normalized entropy",
       linetype = "Covariates",
       title = "Classification quality (entropy) across waves and models")
```